

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

We are given the following :

- Cost to plant the tree : \$ 90 (one-time cost)
- Each year , the tree produces 7 lemons
- Each lemon sells for \$ 1 . 5 0
- Annual cost to water and feed : \$ 3
- We need to find out **after how many years** Carlos starts **earning money**, meaning when the **annual profit** becomes **positive**, or when **revenue > cost** for that year .

Step 1 : Calculate annual income from lemons
Each year , the tree produces 7 lemons .

Income from lemons = 7 × \$ 1 . 5 0 = \$ 10 . 5 0

Step 2 : Subtract annual cost
Annual cost to water and feed = \$ 3

So , **net annual profit** = \$ 10 . 5 0 - \$ 3 = \$ 7 . 5 0

Even though Carlos is earning \$ 7 . 5 0 each year **after** the initial \$ 90 cost , it is only **after the \$ 90 investment** that we consider whether he starts **earning money** .

But the question is : **How many years until he starts earning money ?**

That is , **when is the accumulated profit (after costs) positive ?**

So we need to compute the **total profit after n years**, minus the initial cost .

- > Total money earned after n years :
- > Revenue = 7 × \$ 1 . 5 0 × n = 10 . 5 n
- > Total cost = \$ 90 (one-time) + \$ 3 n (annual water cost)
- > Net profit after n years = 10 . 5 n - 90 - 3 n = 7 . 5 n - 90

We want to find the smallest integer n such that net profit > 0 .

So :

$$\begin{aligned} 7.5n - 90 &> 0 \\ 7.5n &> 90 \\ n &> 90 / 7.5 \\ n &> 12 \end{aligned}$$

So after **more than 12 years**, the net profit becomes positive .

So at **year 12** :

Total net profit = 7 . 5 × 12 - 90 = 90 - 90 = 0 ⇒ **breakeven**

So after **13 years**, the net profit is **positive** ⇒ he starts **earning money**

Thus , he starts earning money in the **13th year** .

∴ Final answer :

< answer > 13 </ answer >